Richard Larocque

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SKILLS

Proficient in C, C++, Python, Java, Haskell
Familiar with Perl, HTML, JavaScript, Go
Deep understanding of Linux systems
In-depth knowledge of TCP/IP specifications
Exprience with software projects large and small
Bilingual (English and French)

WORK EXPERIENCE

Software Engineer — August 2011 — August 2014 (Chrome)
Google Inc., Seattle, WA — August 2014 — Present (Infrastructure)

- Helped to maintain and improve Chrome's Sync feature
- Learned to work on software that sees over 100 commits per day
- Devised an algorithm for representing list orderings in NoSQL data stores
- Used OLAP to transform server logs into actionable performance metrics
- Worked with Docker and Kubernetes container technologies

Design Engineer — May 2009 — July 2011 (full-time)
Avvasi Inc., Waterloo, ON — September 2008 — December 2008 (intern)

- Implemented code to analyze GTP traffic in a live network
- Designed a product licensing solution based on digital signatures
- Improved inter-thread messaging performance with a set of lockless queues
- Developed systems for optimized video delivery over mobile networks
- Helped design a backend for recording and visualizing network status in real time
- Watched the company grow from 10 to 50 employees

System Software Engineer — January 2008 — May 2008
Nvidia Corporation, Santa Clara, California

- Developed tests to measure power management performance on Mac OS X
- Helped implement a content protection system based on X.509 certificates
- Worked on the maintenance of Nvidia's Mac OS X drivers

Developer — May 2007 — August 2007
Wind River Systems, Ottawa, Ontario

- Integrated patches into Wind River's Linux kernel
- Reduced Wind River Linux's storage footprint

Linux Software Engineer — September 2006 — December 2006
Netsweeper Inc., Guelph, Ontario

- Implemented efficient user name hashing and lookup for a content-filtering server
- Reverse-engineered a set of TCP protocols used to communicate with hardware firewalls and implemented them in Netsweeper's software
- Wrote a PHP-based web interface for collecting and visualizing SNMP statistics

Computer Scientist — January 2006 — April 2006
Environment Canada, Dorval, Quebec

- Ported an application for analyzing precipitation data from IRIX64 to Linux
- Fixed bugs in a surface model related to a change from 64-bit precision to 32-bit

Model Programmer and Developer May 2005 — August 2005
Environment Canada, Saskatoon, Saskatchewan

- Improved the integration between two numerical weather models
- Built an interactive website to allow hydrologists easy access to the latest weather model output

OTHER PROJECTS

- Wrote a Python plugin for XBMC to stream Open Yale Courses content
- Built a Latin grammar quiz site using Haskell, Node.js, and data from Wiktionary
  See: `http://discere.r6l7.com`
- Built a login-free voice chat service using Node.js, WebRTC, and AngularJS
  See: `http://vox.r6l7.com`
- Read more on GitHub: `https://github.com/richardlarocque`
- Completed the January 2014 offering Cryptography I on Coursera

EDUCATION

Bachelor of Software Engineering Graduated in 2009
University of Waterloo
Waterloo, Ontario, Canada

COURSE PROJECTS

- As part of a team of three, used C++ to implement a compiler for a subset of Java 1.3 targeting x86 assembly
- Wrote an interpreter for a subset of ML supporting Hindley-Milner type inference
- Implemented a one instruction, Turing-complete CPU in Verilog
- Worked with a small team to implement a simple RTOS for the Freescale ColdFire
- Developed a Haskell program to solve the travelling salesman problem and wrote a report comparing its performance and SMP scalability against a competing C implementation

HOBBIES & INTERESTS

Rock climbing, marathon running, reading, history, Latin, micro-brewing